

ANDISA KHUMALO

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PROFESSIONAL PROFILE

Final-year Electrical & Computer Engineering student and 3rd-best degree finalist with hands-on expertise in autonomous systems, PLC control, and full-stack development. Built an ML-driven autonomous vehicle, PLC HVAC system, and AZURE deployed web app. Seeking a graduate engineer-in-training role to solve practical technical challenges.

PROFESSIONAL ACTIVITIES & ACHIEVEMENTS

South African Institute of Electrical Engineers (SAIEE)

2024 – Present

Student Member

- Attended technical talks and industry workshops, deepening knowledge of electrical engineering practices and emerging industry trends.
- Networked with guest speakers, including professional engineers and mentors, building a professional network and gaining practical career insights.

SPETM Conference

2026

Presenter

- Presented a research poster on the design and implementation of an electric and autonomous vehicle at the international SPETM Conference (University of Johannesburg), attended by researchers from around the world.
- Engaged with international researchers and industry professionals, explaining the vehicle's system architecture, control strategy, and machine-learning workflow while answering technical questions.

EDUCATION

University of Zululand

2023-Nov 2026

Bachelor of Engineering, Electrical and Computer Engineering

- Selected study areas: engineering systems design, software engineering, assembly language and operating systems, industrial engineering law, occupational health and safety, power systems, signals and systems and communications & networks.

Qhamuka High School

2021

Secondary School (Matric)

PRACTICAL INVOLVEMENTS & CERTIFICATIONS

1. Eskom Espo program

Volunteer

- Evaluated approximately 10 engineering, electronics, robotics, coding and science projects, assessing technical functionality and project presentations against the programme's criteria.
- Assessed projects, questioned participants on their technical work, provided feedback and contributed to the grading and selection process.
- Presented assessment results with the programme team; one of the candidates assessed achieved 2nd position.

2. OpenEuler certification (Linux systems administration), Huawei ICT Academy

Certified in Linux system administration using OpenEuler, covering core Linux operating system management, configuration, and administration.

3. OpenGauss certification (database administration), Huawei ICT Academy

Certified in OpenGauss database administration, covering database management, configuration, and fundamental administration practices.

TECHNICAL SKILLS

Programming: C#, Python, C/C++, MATLAB, Arduino programming, Kotlin/Android development

Industrial Automation: Siemens PLC programming, ladder logic, SCADA concepts, PID control, Instrumentation

Software & Web: ASP.NET Core MVC, HTML/CSS, SQL server, EF Core, GIT/GITHUB

Embedded Systems: Raspberry Pi, Arduino, STM32, serial communication, sensor & camera integration, motor control

Computer Vision & AI: OpenCV, lane keeping, machine learning workflows, camera data collection & model deployment

Control & Simulation: MATLAB/Simulink, PID design, modelling and engineering simulation

Engineering Documentation: IEEE-format technical reports, system architecture documentation, database administration

ENGINEERING & TECHNOLOGY

Autonomous Electric Vehicle - Final-Year Capstone

2026

- Currently designing and building a low-cost autonomous electric vehicle integrating a Raspberry Pi 4, Arduino Mega and a single drive motor powered through a 12V-to-5V buck-converter power system.
- Developed a machine-learning workflow to train the vehicle to follow lane lines from onboard camera data, including data collection, labelling and model deployment.
- Integrated embedded control, two ultrasonic sensors, camera vision and 4G connectivity into a multidisciplinary engineering system.
- Implementing real-time closed-loop steering and obstacle-aware control, with an Android monitoring concept and Raspberry Pi backend.

AK Mechanics - Full-Stack Vehicle Fleet Management System

2026

- Built a full-stack web application using ASP.NET Core MVC, Entity Framework Core and SQL Server.
- Implemented role-based access control for clients, service providers and administrators using ASP.NET Identity.
- Deployed the application using Microsoft Azure App Service and Azure SQL Database.

Smart Airport Luggage Tracking System - Capstone Group Project

2026

- Developed the proposed tracking architecture covering RFID identification, GPS positioning, IoT sensing, communication between luggage and the central system, database integration and web/app interfaces.
- Designed the proposed tracking and communication approach around RFID, LoRa and MQTT, including the tracking algorithm and sensor subsystem.
- Contributed to the system proposal and execution plan; project outputs were presented as a proposed solution rather than a deployed system.

REFERENCES

Dr OE Oni from University of Zululand

- onio@unizulu.ac.za

Mr. B Khoza from University of Zululand

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